

13 CAN Communication

The CAN communication is based on the SAE J1939 protocol. The hardware is supported as in the protocol and messages are using the open addresses 0x18FFxxxx.

13.1 Standard CAN communication

Standard CAN communication is used to broadcast most important data on the CAN bus
128bit 125kbps ->976msg/s

All values reaching 2 or more bytes are always send with MSB first.

Source	ID- start	ID- end	Message	Rate (Msg/s)	Bus %
	0x18FF00xx		Battery Status	10	1.02
	0x18FF01xx		DC input and Output	1	0.102
	0x18FF03xx		Operating State and status bitmaps	1	0.102
	0x18FF04xx		Function Operating state and failure level	1	0.102
	0x18FF05xx		Failure code buffer	1	0.102
	0x18FF06xx		Internal and Cell Temperature	1	0.102
	0x18FF07xx		Serial number and Manufacture Date	1	0.102
	0x18FF08xx		Firmware and Hardware Revision	1	0.102
	0x18FF09xx		AC Input and AC Output	1	0.102
	0x18FF10xx		Cell voltage	1	0.102
	0x18FF18xx		Cell SOC	1	0.102
	0x18FF20xx		Jumpstart & Solar	1	0.102
	0x18FF21xx		IO Input Voltages	1	0.102
	0x18FF30xx		System Information	1	
	0x18FF31xx		System energy Meter 1	0,01667	
	0x18FF32xx		System energy Meter 2	0,01667	
	0x18FF40xx		G3 Combi	1	
Total				20	2.04%

The maximum number of units connected in one network is: 40pcs.

13.1.1 0x18FF00xx – Battery Status – OK

The LPS calculates SOC and measure current and cell voltages and transmits them over CAN.

The SOC Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF00xx	SOC		Battery Current		Remaining Time		Battery Ah	

- SOC is the pack state of charge. It shall be unsigned 16-bits, with a resolution of 1/65535 (%) per bit, and a range of 0% to 100%.
- Battery Current. Resolution is 100mA Signed integer.
- Remaining Time, is how many minutes until battery is empty or >97%SOC, Signed integer - negative = discharge time.
- Battery Ah is the amount of Ah available in the battery. Resolution is 0.1A
- Transmit period: 100 milliseconds.

13.1.2 0x18FF01xx – DC Input and Output – OK

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN.

The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF01xx	DC Input Voltage		DC Input Current		DC Output Voltage		DC Output Current	

- DC Input Voltage is voltage measured before the DC/DC converters. Resolution is 1mV.
- DC Input Current. Resolution is 10mA Signed integer.
- DC Output Voltage is voltage measured at the Output connector. Resolution is 1mV.
- DC Output Current. Resolution is 10mA Signed integer.
- Transmit period: 1 second.

13.1.3 0x18FF03xx – Operating State and status bitmaps – OK

The LPS Operating State and status bitmaps Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF03xx	Operating State	Failure Level	Test mode	Wakeup Flags (LSB First)		LPS Outputs	Cell Count	Temperature Sensor Count

- a. LPS Operating State values is a specified in the Operating state diagram.

Value	Bit Name	Description
0	Initial Power up	
1	Final Power Down	
2	Sleep	
3	Control Circuits Power Up	
4	Control Circuits Power Down	
5	Control Circuits On	
6	Power Supply, Power up	
7	Power Supply, Power down	
8	Power Supply, On	
FF	BIOS Test mode	

- b. LPS Failure level is a specified in: Failure levels:

Value	Bit Name	Description
0	OK	
1	Warning	
2	Simple failure	
3	Empty	
4	Critical Failure	

- a. LPS Test mode

Value	Bit Name	Description
0	Normal mode	
2	BIOS test mode	
3	Application test mode	
4	No Protection test mode	

- b. LPS Wakeup flags

Bit #	Bit Name	Description
0	Mains Wakeup	
1	Reserved	
2	Engine Wakeup (D+, Clamp 15)	
3	Engine Ripple Wakeup	
4	Wakeup blocked	
5	Reserved	
6	Power Button 230VAC	
7	Power Button 12VDC	
8	IO Terminal Wakeup	
9	IO 1 Wakeup (Remote)	
10	IO 2 Wakeup (Data)	
11	IO 3 Wakeup (Front Data)	
12	Reserved	
13	Reserved	
14	Reserved	
15	Reserved	

- a. LPS Outputs:

Bit #	Bit Name	Description
0	IO 1	
1	IO 2	
2	IO 3	
3	IO Terminal	
4	DC/DC Wakeup	
5	DC Relay	
6	PE Relay	

Bit #	Bit Name	Description
7	Transfer Relay	

- b. Cell count, is a counter of the number of cells inside the battery.
- c. Temperature sensor count, is a counter of the number of Cell temperature sensors inside the battery.
- d. Transmit period: 1 second.

13.1.4 0x18FF04xx – Function Operating State – OK

The battery reports information about operating state and failure levels in the battery.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF04xx	Inverter Operating State	Inverter Failure Level	Charger Operating State	Charger Failure Level	DC Input Operating State	DC Input Failure Level	DC Output Operating State	DC Output Failure Level

- a. Inverter, Charger, DC Input and DC Output Operating State

Value	Bit Name	Description
0	OFF	
1	Wakeup	
2	Ready To Start	
3	Starting	
4	Stopping	
5	ON	

- b. Inverter, Charger, DC Input and DC Output Failure Level

Value	Bit Name	Description
0	OK	
1	Warning	
2	Simple failure	
3	Empty	
4	Critical Failure	

- c. Transmit period: 1 second.

13.1.5 0x18FF05xx – Failure Code Buffer – OK

The Failure Code Buffer Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF05xx	FC_0	FC_1	FC_2	FC_3	FC_4	FC_5	FC_6	FC_7

- a. FC_0 to FC_7 is actual failures in the LPS. If values are 'Zero' meaning no failure. Failure codes can be found in failure code table.
- b. Transmit period: 1 second.

13.1.6 0x18FF06xx – LPS Internal and Battery temperature – OK

The LPS report internal temperature of hardware

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF06xx	Internal Temperature 1		Internal Temperature 2		Internal Temperature 3		Cell Temperature AVG	

- a. Internal Temperature 1 – Trafo temperature (Power Board)
- b. Internal Temperature 2 – IGBT temperature (Power Board)
- c. Internal Temperature 3 – DC/DC Converter temperature
- d. Resolution of all temperatures is 1/256 °C/bit.
- e. Cell Temperature AVG is the average cell temperature.
- f. All temperatures shall be signed 2's complement, 16 bits, MSB first, with a resolution of 0.0039 degree C per bit. (1/256 °C/bit)
- g. If a thermistor input is physically disconnected from the LPS, the LPS report a value of 0x7FFE for the corresponding temperature.
- h. Transmit period: 1 second.

13.1.7 0x18FF07xx – Serial Number and Manufacture Date – OK

The LPS reports the manufacture date and serial number over CAN. Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF07xx	Year	Month	Day	Batch Number			Serial Number	

- Year shall be unsigned 8 bits with MSB first with valid values from 0-99 decimal.
- Month shall be unsigned 8 bits with MSB first with valid values from 0-12 decimal.
- Day shall be unsigned 8 bits with MSB first with valid values from 0-31 decimal. In cases where Day is not used, this value shall be 0xFF.
- Batch Number is number of manufacture batch of BMS. Batch number consists of 3 bytes each can have the value 0-99 which gives a batch number of 6 digits.
- Serial number is the serial number of the BMS. The serial number consists of 2 bytes each can have the value 0-99 which gives a serial number of 4 digits.
- Transmit period: 1 second.

13.1.8 0x18FF08xx – Firmware and Hardware Revision – OK

The LPS reports the software revision over CAN. The Software Revision Message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF08xx	FW Major Revision	FW Minor Revision	FW Build Revision	HW Major Revision	HW Minor Revision			Cell Balance bitmap

- The FW Major Revision
- The FW Minor Revision
- The FW Build Revision
- The HW Major Revision. It is hardware revision.
- The HW Minor Revision. It is hardware revision.
- Cell balance bitmap

Bit #	Bit Name
7	Not used
6	Not used
5	Not used
4	Not used
3	Cell 4
2	Cell 3
1	Cell 2
0	Cell 1

- Transmit period: 1 second.

13.1.9 0x18FF09xx – AC Input and AC Output – OK

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN.

The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF09xx	AC Input Voltage		AC Input Current		AC Output Voltage		AC Output Current	

- AC Input Voltage is voltage measured before the transfer relay Resolution is 100mV.
- AC Input Current is measured after the transfer relay. Resolution is 1mA unsigned integer.
- AC Output Voltage is measured at the Output connector. Resolution is 100mV.
- DC Output Current is measured at the Output connector. Resolution is 10mA unsigned integer.
- Transmit period: 1 second.

13.1.10 0x18FF10xx – Cell Voltage Message – OK

The LPS shall have the ability to transmit all cell voltages on the CAN bus.

This message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF10xx	Cell 1		Cell 2		Cell 3		Cell 4	

- Cell 1 is the voltage of the first cell (at the negative battery terminal).
- Cell voltages shall be unsigned 16 bits, MSB first, with a resolution of 1/8192 volts (i.e. 122 μ V) per bit.
- If a cell voltage sense line is physically disconnected from the battery, the MCU shall report a value of 0xFFFE for that cell voltage.
- Transmit period: 1 second.

13.1.11 0x18FF18xx – Cell SOC Message – OK

The LPS shall have the ability to transmit all cell SOC on the CAN bus.

This message is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF18xx	SOC Cell 1		SOC Cell 2		SOC Cell 3		SOC Cell 4	

- Cell 1 is the SOC of the first cell (at the negative battery terminal).
- Cell SOC shall be unsigned 16 bits, MSB first, with a resolution of 1/65535 % (i.e. 0,00001529%) per bit.
- Transmit period: 1 second.

13.1.12 0x18FF20xx – Jumpstart & Solar – OK

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN.

The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF20xx	Jumpstart Time		IO Terminal		Solar Current		Solar Operating State	Solar Failure Level

- Jumpstart time is the remaining time of jumpstart charge. Resolution is 10msec.
- IO Terminal voltage (Solar input voltage) has resolution of 1mV.
- Solar Current is the current into the battery. Resolution is 1mA Signed integer.
- All Values are transmitted with MSB byte first.
- Transmit period: 1 second.

13.1.13 0x18FF21xx – IO Input Voltage – OK

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN.

The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF21xx	IO 1 Voltage		IO 2 Voltage		IO 3 Voltage		IN Terminal	

- IO 1, IO 2, IO 3 and IN Terminal voltages do all have resolution of 1mV.
- All Values are transmitted with MSB byte first.
- Transmit period: 1 second.

13.1.14 0x18FF22xx – AC Power

The LPS measures AC Input and AC Output power, and transmits them over CAN.

The AC Input and AC Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF22xx	AC Input Power		AC Output Power		0x00		0x00	

- AC Input Power and AC Output Power do all have resolution of 1W.
- All Values are transmitted with MSB byte first.
- Transmit period: 1 second.

13.1.15 0x18FF23xx – E-GEN III – Terminal Temperatures

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN. The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF23xx	Terminal A		Terminal B		0x0000		0x0000	

- a. All temperatures shall be signed 2's complement, 16 bits, MSB first, with a resolution of 0.0039 degree C per bit. (1/256 °C/bit)

Power Module	Location	Terminal A	Terminal B
Master	0x01	Battery Terminal A	Battery Terminal B
Slave 240	0x03	DC Input Terminal A	DC Input Terminal B
Single 1	0x11	DC Output Terminal A	DC Output Terminal B

- b. All Values are transmitted with MSB byte first.
c. Transmit period: 1 second.

13.1.16 0x18FF30xx – System Information

The LPS receive information from all units in the system and calculate some system information. These information's are broadcasted when other components are found on the CAN communication. If the system consists of multiple LPS's will only the master (unit with highest serial number) broadcast the message.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF30xx	SOC							

- a. SOC is the State Of Charge for the entire system. Resolution is 1/65535 (%) per bit, and a range of 0% to 100%.
b. All Values are transmitted with MSB byte first.
c. Transmit period: 1 second.

13.1.17 0x18FF31xx – System Energy Meter 1

The LPS receive information from all units and calculate input and output power as an energy meter. If the system consists of multiple LPS's will only the master (unit with highest serial number) broadcast the message.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF31xx	AC Input Power kWh				AC Output Power kWh			

- a. AC Input Power is the amount of kWh consumed from 230vac mains. Resolution is 1/65535 per bit, and a range of 0 to 65535,99995 kWh.
b. AC Output Power is the amount of kWh delivered on both 230vac and 400vac. Resolution is 1/65535 per bit, and a range of 0 to 65535,99995 kWh.
c. All Values are transmitted with MSB byte first.
d. Transmit period: 60 second.

13.1.18 0x18FF32xx – System Energy Meter 2

The LPS receive information from all units and calculate input and output power as an energy meter. If the system consists of multiple LPS's will only the master (unit with highest serial number) broadcast the message.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF32xx	DC Input Power kWh				Solar Power kWh			

- e. DC Input Power is the amount of kWh charged from vehicle. Resolution is 1/65535 per bit, and a range of 0 to 65535,99995 kWh.
f. Solar Power is the amount of kWh charged from solar panels. Resolution is 1/65535 per bit, and a range of 0 to 65535,99995 kWh.
g. All Values are transmitted with MSB byte first.

- h. Transmit period: 60 second.

13.1.19 0x18FF40xx – G3 Combi Information

The LPS receive information from all units in the system and calculate some average values for the G3 Combi's. These information's are broadcasted when G3 Combi's are found on the single wire communication. If the system consists of multiple LPS's will only the master (unit with highest serial number) broadcast the message.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF30xx	DC Current		AC Input Voltage		AC Input Power		AC Output Power	

- a. DC Current is average for all the connected G3 Combi's. Resolution is 1/10 (A) per bit, and a range of -3267,8 – 3267,7A.
- b.
- c. All Values are transmitted with MSB byte first.
- d. Transmit period: 1 second.

13.1.20 0x18FFFFxx – Time / Sync

The LPS measures DC Input and DC Output current and voltage, and transmits them over CAN.

The DC Input and Output is defined as follows:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18FF21xx	TimeStamp				TXCorrection		0x00	0x00

- a. TimeStamp is a counter of 32bit that rolls over. Time is incremented ever 50ysec. (Res = 50ysec)
- b. TXCorrection is a correction value to last time message since TimeStamp was captured until TX interrupt is generated. (Res = 50ysec)
- c. All Values are transmitted with MSB byte first.
- d. Transmit period: 100 msec.

13.1.21 0x19FF00xx – Serial Number and Manufacture Date

Reports unit serial number and date of manufacture.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF00xx	Year	Month	Day	Batch Number			Serial Number	

- a. Manufacture Date consists of 3 bytes with values from 0-99, 0-12 and 0-31 decimal. Date Format: DD:MM:20YY
- b. In cases where Date is not used, values will be either 0x00 or 0xFF.
- c. Batch Number is number of manufacture batch of LPS. Batch number consists of 3 bytes each can have the value 0-99 which gives a batch number of 6 digits. Format: 000000.
- d. Serial number is the serial number of the LPS. The serial number consists of 2 bytes each can have the value 0-99 which gives a serial number of 4 digits. Format: 0000
The Unit serial number uses both Batch number and Serial number. Format: 000000-0000
- e. Transmit period: After Powerup and on request

13.1.22 0x19FF01xx – Second Serial Number and Firmware update date

Reports the second serial number and date for last firmware update.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF01xx	Year	Month	Day	Batch Number			Serial Number	

- Last Firmware Update Date consists of 3 bytes with values from 0-99, 0-12 and 0-31 decimal.
Date Format: DD:MM:20YY
- In cases where Date is not used, values will be either 0x00 or 0xFF.
- Batch Number is number of manufacture batch of LPS. Batch number consists of 3 bytes each can have the value 0-99 which gives a batch number of 6 digits. Format: 000000.
- Serial number is the serial number of the LPS. The serial number consists of 2 bytes each can have the value 0-99 which gives a serial number of 4 digits. Format: 0000
The Unit serial number uses both Batch number and Serial number. Format: 000000-0000
- Transmit period: After Powerup and on request

13.1.23 0x19FF02xx – Firmware and Hardware Revision

Reports the software and hardware revisions over CAN.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF02xx	FW Major Revision	FW Minor Revision	FW Build Revision	HW Major Revision	HW Minor Revision	FW2 Major Revision	FW2 Minor Revision	FW2 Build Revision

- FW Revision consists of 3 bytes with each value from 00-99 decimal. Format: 00:00:00
- HW Revision consists of 2 bytes with each value from 00-99 decimal. Format: 00:00
- FW2 Revision consists of 3 bytes with each value from 00-99 decimal. Format: 00:00:00
- Transmit period: After Powerup and on request

13.1.24 0x19FF03xx & 0x19FF04xx – Clayton Power Part number

Reports the CP Part number over CAN. The Part number consists of 16 chars.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF03xx	Char n	Char n+1	Char n+2	Char n+3	Char n+4	Char n+5	Char n+6	Char n+7
0x19FF04xx	Char n+8	Char n+9	Char n+10	Char n+11	Char n+12	Char n+13	Char n+14	Char n+15

- Part number is at string (ASCII) of 16 characters.
- If part number is shorter than 16 characters empty bytes will have value '0'
- Transmit period: After Powerup and on request

13.1.25 0x19FF05xx - 0x19FF08xx – Clayton Power Part Description

Reports the CP Part Description over CAN. The Part Description consists of 32 chars.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19FF05xx	Char n	Char n+1	Char n+2	Char n+3	Char n+4	Char n+5	Char n+6	Char n+7
0x19FF06xx	Char n+8	Char n+9	Char n+10	Char n+11	Char n+12	Char n+13	Char n+14	Char n+15
0x19FF07xx	Char n+16	Char n+17	Char n+18	Char n+19	Char n+20	Char n+21	Char n+22	Char n+23
0x19FF08xx	Char n+24	Char n+25	Char n+26	Char n+27	Char n+28	Char n+29	Char n+30	Char n+31

- Part description is at string (ASCII) of 32 characters.
- If part description is shorter than 32 characters empty bytes will have value '0'
- Transmit period: After Powerup and on request

13.1.26 0x18EAxxxy – Request PGN

Messages can be requested from the LPS by sending a message directly to the LPS. Afterwards do the LPS respond by broadcasting the requested message or series of messages.

Request message:

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x18EAxxyy	Message ID			0x00	0x00	0x00	0x00	0x00

Device ID = 0xFF (Global address)

Host (PC) = 0xFE

,

13.1.27 0x19EFxxyy – Power Control: 12VDC Output

The 12VDC Output state can be controlled via CAN communication by sending a message or status read, by sensing a request message. The LPS will respond with a similar message when requested or when set.

Address xx = Device address (LPS address) (Global address: 0xFF)

Address yy = Host address (Controlling unit) (Address cannot be 0x00 or 0xFF)

Power Control 12VDC Output C Read

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x40	0x00	0x9F	0x00*			

*Response value On=65536 / Off=0 (LSB First)

Power Control 12VDC Output Set (On/Off)

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x41	0x00	0x9F	On=65536 / Off=0 (LSB First)			

13.1.28 0x19EFxxyy – Power Control: 230VAC Output (Inverter)

The 230VAC Output state can be controlled via CAN communication by sending a message or status read, by sensing a request message. The LPS will respond with a similar message when requested or when set.

Address xx = Device address (LPS address) (Global address: 0xFF)

Address yy = Host address (Controlling unit) (Address cannot be 0x00 or 0xFF)

Power Control 230VAC Output Read

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x40	0x00	0x9E	0x00*			

*Response value On=65536 / Off=0 (LSB First)

Power Control 230VAC Output Set (On/Off)

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x41	0x00	0x9E	On=65536 / Off=0 (LSB First)			

13.1.29 0x19EFxxyy – Power Control: Jump Start

The Jump Start state can be controlled via CAN communication by sending a message or status read, by sensing a request message. The LPS will respond with a similar message when requested or when set.

Address xx = Device address (LPS address) (Global address: 0xFF)

Address yy = Host address (Controlling unit) (Address cannot be 0x00 or 0xFF)

Power Control Jump Start Read

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x40	0x01	0x01	0x00*			

*Response value is the time left until Jump Start is done. (0=off / 65536 = 1 hour)

Power Control Jump Start Set (On/Off)

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x41	0x01	0x01	On=65536 / Off=0 (LSB First)			

13.1.30 0x19EFxxyy – Energy Meter (Read)

The energy meters can be read via CAN communication by sending a message. The LPS will respond with a similar message when requested.

Address xx = Device address (LPS address) (Global address: 0xFF)

Address yy = Host address (Controlling unit) (Address cannot be 0x00 or 0xFF)

Message to read energy meter.

CAN-ID	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
0x19EFxxyy	0x00	0x40	0x00	ID	0x00			

- a) ID reference to type of value requested.

ID	Patermeter
140	AC Input Charge kWh
142	DC Input Charge kWh
144	DC Output Charge kWh
146	Solar Charge kWh

- b) Response value is on kWh and the value is fixed point where the first 16bit is two complement and the last 16bit is fragment. (Divide with 65536 to get decimal number.)